

Part I - Theory

Exercise 1

- (a) We will generally denote the column of the class weight matrix corresponding to the outcome b_i as $\underline{x}_{b_i}$. Now, we first derive the likelihood:

$$L(X) = P(b_1, \dots, b_n | \underline{a}_1, \dots, \underline{a}_n) = \prod_{i=1}^n P(b_i | \underline{a}_i) = \prod_{i=1}^n \frac{\exp(\underline{a}_i^T \underline{x}_{b_i})}{\sum_k \exp(\underline{a}_i^T \underline{x}_k)}$$

Taking the log we get

$$\begin{aligned} \log L(X) &= \log \left\{ \prod_{i=1}^n \frac{\exp(\underline{a}_i^T \underline{x}_{b_i})}{\sum_k \exp(\underline{a}_i^T \underline{x}_k)} \right\} = \sum_{i=1}^n \log \left\{ \frac{\exp(\underline{a}_i^T \underline{x}_{b_i})}{\sum_k \exp(\underline{a}_i^T \underline{x}_k)} \right\} = \\ &= \sum_{i=1}^n \left[\underline{a}_i^T \underline{x}_{b_i} - \log \sum_k \exp(\underline{a}_i^T \underline{x}_k) \right] \end{aligned}$$

Negating the expression gives the desired result. $\square$

- (b) We wish to show that $g: \underline{u} \mapsto \log \sum_{k=1}^d e^{u_k}$ is convex on $\mathbb{R}^d$.
 $g: \mathbb{R}^d \rightarrow \mathbb{R}$

For g to be convex on $\mathbb{R}^n$ we need

$$g(\alpha \underline{u}_1 + (1-\alpha) \underline{u}_2) \stackrel{?}{\leq} \alpha g(\underline{u}_1) + (1-\alpha) g(\underline{u}_2) \quad \forall \underline{u}_1, \underline{u}_2 \in \mathbb{R}^n, \forall \alpha \in [0, 1]$$

$$\log \sum_{k=1}^d \exp(\alpha u_{1,k} + (1-\alpha) u_{2,k}) \stackrel{?}{\leq} \alpha \log \sum \exp(u_{1,k}) + (1-\alpha) \log \sum \exp(u_{2,k})$$

Exponentiating both sides,

$$\sum_k \exp(\alpha u_{1,k} + (1-\alpha) u_{2,k}) \stackrel{?}{\leq} \exp \left\{ \log \left[\sum \exp(u_{1,k}) \right]^\alpha + (1-\alpha) \log \left[\sum \exp(u_{2,k}) \right]^{1-\alpha} \right\}$$

$$\sum_k \exp(\alpha u_{1,k}) \exp((1-\alpha) u_{2,k}) \stackrel{?}{\leq} \left[\sum \exp(u_{1,k}) \right]^\alpha \left[\sum \exp(u_{2,k}) \right]^{1-\alpha}$$

$$\sum_k \frac{\exp(\alpha u_{1,k})}{\left[\sum \exp(u_{1,k}) \right]^\alpha} \cdot \frac{\exp((1-\alpha) u_{2,k})}{\left[\sum \exp(u_{2,k}) \right]^{1-\alpha}} \stackrel{?}{\leq} 1$$

$$\sum_k P(b_i = k | \underline{u}_1)^\alpha P(b_i = k | \underline{u}_2)^{1-\alpha} \stackrel{?}{\leq} 1 \quad \text{recognising the Softmax pmf}$$

$$\sum_k P(b_i = k | \underline{u}_1)^\alpha \frac{P(b_i = k | \underline{u}_1)^{1-\alpha}}{P(b_i = k | \underline{u}_1)^{1-\alpha}} P(b_i = k | \underline{u}_2)^{1-\alpha} \stackrel{?}{\leq} 1$$

$$\sum_k P(b_i = k | \underline{u}_1) \frac{P(b_i = k | \underline{u}_2)^{1-\alpha}}{P(b_i = k | \underline{u}_1)^{1-\alpha}} \stackrel{?}{\leq} 1$$

$$E_{b_i | \underline{u}_1} \left\{ \left[\frac{P(b_i = k | \underline{u}_2)}{P(b_i = k | \underline{u}_1)} \right]^{1-\alpha} \right\} \stackrel{?}{\leq} 1$$

By Jensen's inequality $E_{b_i | \underline{u}_1} \left\{ \left[\frac{P(b_i = k | \underline{u}_2)}{P(b_i = k | \underline{u}_1)} \right]^{1-\alpha} \right\} \leq \left\{ E_{b_i | \underline{u}_1} \left[\frac{P(b_i = k | \underline{u}_2)}{P(b_i = k | \underline{u}_1)} \right] \right\}^{1-\alpha}$

This is because $y = x^\theta$ is concave by the definition of convex function using the positive semi-definiteness of the Hessian (which here reduces to a scalar):

$$\frac{\partial}{\partial x} y = \theta x^{\theta-1}, \quad \frac{\partial^2}{\partial x^2} y = \theta(\theta-1) x^{\theta-2} \leq 0 \quad \text{for any } \theta \in [0, 1]$$

The RHS of the Jensen inequality is

$$\begin{aligned} \left\{ E_{b_i | \underline{u}_1} \left[\frac{\mathbb{P}(b_i = k | \underline{u}_2)}{\mathbb{P}(b_i = k | \underline{u}_1)} \right] \right\}^{1-\alpha} &= \sum_{k=1}^d \frac{\mathbb{P}(b_i = k | \underline{u}_2)}{\cancel{\mathbb{P}(b_i = k | \underline{u}_1)}} \cancel{\mathbb{P}(b_i = k | \underline{u}_1)} = \\ &= \sum_k \mathbb{P}(b_i = k | \underline{u}_2) = 1 \quad \text{since it is a pmf} \end{aligned}$$

This implies that

$$E_{b_i | \underline{u}_1} \left\{ \left[\frac{\mathbb{P}(b_i = k | \underline{u}_2)}{\mathbb{P}(b_i = k | \underline{u}_1)} \right]^{1-\alpha} \right\} \leq 1 \quad \square$$